

Software Requirements Specification (SRS)

Agri-Lend: Agricultural Credit Intelligence Platform

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for the **Agri-Lend** platform — a data-driven agricultural credit intelligence system. This document is intended for developers, data scientists, product managers, and technical stakeholders involved in designing, building, and maintaining the platform.

1.2 Project Overview

Agri-Lend is a digital platform that transforms raw farm data into reliable agricultural credit scores. It bridges the gap between smallholder farmers — who lack traditional collateral such as formal land titles — and financial institutions, which face significant information asymmetry when evaluating agricultural lending risk.

The platform operates as a **Technical Service Provider (TSP)** and intelligence layer positioned between farmers and banks, rather than acting as a lender itself.

1.3 Scope

The Agri-Lend platform encompasses three integrated core components:

Component	Internal Name	Description
Data Ingestion Layer	The Harvester	Satellite API integration, IoT-readiness, and multi-source data collection
ML Intelligence Engine	The Brain	Predictive models for yield forecasting and real-time risk scoring
User-Facing Interfaces	The Marketplace	Bank-facing web dashboard and farmer-facing mobile application

1.4 Definitions and Acronyms

Term	Definition
NDVI	Normalized Difference Vegetation Index — a satellite-derived measure of crop health
GEE	Google Earth Engine — geospatial analysis platform
TSP	Technical Service Provider
Mobile Money	Digital financial services tied to mobile phone accounts (e.g., M-Pesa)
Credit Score	A numerical representation of a farmer's creditworthiness derived from platform data
SaaS	Software as a Service
IoT	Internet of Things — connected sensors and devices
NGO	Non-Governmental Organization
KYC	Know Your Customer — identity verification process
API	Application Programming Interface

1.5 References

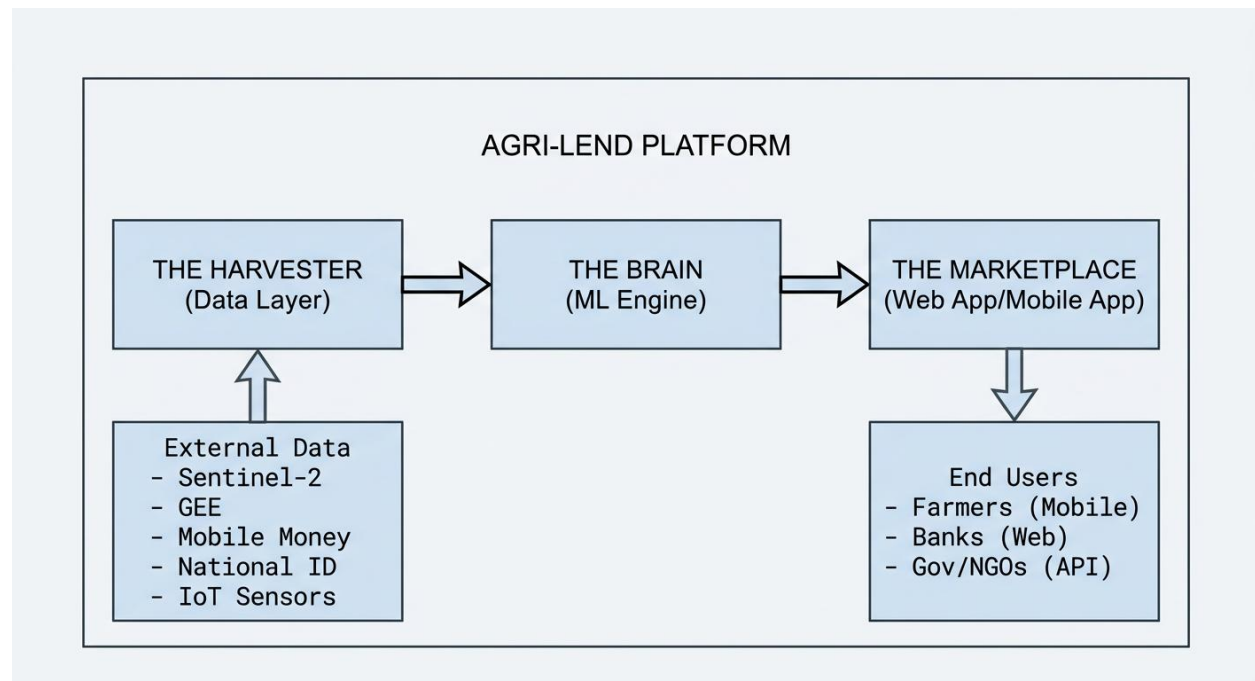
- Agri-Lend Pitch Deck (2024)
- Sentinel-2 Satellite Data Documentation (ESA)
- Google Earth Engine API Documentation
- FastAPI Framework Documentation

2. Overall Description

2.1 Product Perspective

Agri-Lend is a new, standalone platform that integrates with several external systems:

- **Satellite data providers** (Sentinel-2, Google Earth Engine)
- **National Digital ID systems** (for farmer identity verification)
- **Mobile money providers** (for transaction history and loan disbursement)
- **Partner bank systems** (via API for score delivery and monitoring)
- **Agricultural cooperatives** (as data and distribution partners)



2.2 User Classes and Characteristics

2.2.1 Farmers

- **Profile:** Smallholder farmers, potentially with limited digital literacy
- **Access Method:** Mobile application (low-bandwidth optimized)
- **Primary Needs:** Loan access, credit score visibility, farm performance insights, SMS alerts
- **Technical Proficiency:** Low to moderate; multilingual support required

2.2.2 Bank/ Financial Institution Officers

- **Profile:** Credit analysts, loan officers, risk managers at partner banks
- **Access Method:** Web-based dashboard
- **Primary Needs:** Credit score reports, risk heatmaps, portfolio monitoring, new customer discovery

- **Technical Proficiency:** Moderate to high

2.2.3 Platform Administrators

- **Profile:** Agri-Lend internal staff
- **Access Method:** Admin web portal
- **Primary Needs:** User management, data pipeline monitoring, model performance tracking, partner onboarding
- **Technical Proficiency:** High

2.2.4 Government & NGO Analysts (Future)

- **Profile:** Policy researchers, development organizations
- **Access Method:** API or downloadable reports
- **Primary Needs:** Regional productivity data, aggregated lending trends, food security indicators

2.3 Assumptions and Dependencies

- Sentinel-2 and Google Earth Engine APIs remain publicly accessible and maintain current data cadence
- Partner banks have existing digital infrastructure capable of consuming REST API responses
- Target farmers possess or can obtain mobile phones capable of running the mobile application
- National digital ID systems expose integration APIs for identity verification
- Mobile money providers (e.g., M-Pesa, Telebirr) offer data-sharing APIs for transaction history
- Internet connectivity, while intermittent, is available in target deployment regions

2.4 Constraints

- Must comply with applicable data protection and financial regulations in operating countries
- Initial pilot is focused on high-value crops: **Coffee and Teff**
- Satellite revisit time (Sentinel-2: approximately 5 days at equator) limits real-time geospatial update frequency
- Platform must operate effectively in low-bandwidth environments for farmer-facing features

3. System Architecture Overview

3.1 Technology Stack

Layer	Technology	Purpose
Backend	Python, FastAPI, PostgreSQL	Core API services, business logic, data persistence
Frontend (Web)	React (Vite)	Bank dashboard and admin portal
Frontend (Mobile)	Mobile App (React Native or Flutter — TBD)	Farmer-facing mobile application
Data Science	Python-based ML models (scikit-learn, TensorFlow, or equivalent)	Credit scoring, yield prediction, risk classification
Geospatial / Satellite	Sentinel-2, Google Earth Engine	Remote crop health monitoring (NDVI and related indices)
Database	PostgreSQL with PostGIS extension	Spatial data storage and querying
Messaging / Alerts	SMS Gateway integration	Farmer SMS notifications

3.2 Deployment Architecture

- Cloud-hosted infrastructure (scalable, containerized deployment recommended via Docker/Kubernetes)
- RESTful API-first design to enable bank integrations and future third-party access
- Separate environments: Development, Staging, Production

4. Functional Requirements

This section is organized by the three core platform components, with additional modules for cross-cutting features.

MODULE 1: THE HARVESTER — Data Ingestion Layer

FR-H-001: Satellite Data Ingestion

Attribute	Detail
ID	FR-H-001
Title	Satellite Crop Health Data Ingestion
Description	The system shall automatically fetch and process satellite imagery for registered farm parcels using the Sentinel-2 dataset via Google Earth Engine.
Inputs	Farm GPS coordinates / polygon boundary, date range
Processing	Calculate NDVI and other vegetation indices per parcel
Outputs	Time-series NDVI values stored per farm profile
Frequency	Automated fetch aligned to Sentinel-2 revisit cycle (~5 days)
Priority	HIGH

FR-H-002: Transactional Data Ingestion

Attribute	Detail
ID	FR-H-002
Title	Sales and Payment History Ingestion
Description	The system shall ingest farmer sales records and payment history from connected cooperative systems or manual farmer input via the mobile app.
Inputs	Sales records (date, crop type, quantity, price), payment records
Outputs	Structured transactional history linked to farmer profile
Priority	HIGH

FR-H-003: Mobile Money Data Ingestion

Attribute	Detail
ID	FR-H-003
Title	Mobile Money Transaction History Integration
Description	The system shall integrate with mobile money provider APIs (with farmer consent) to retrieve transaction history for use as alternative credit data.
Inputs	Farmer's mobile money account identifier, authorized access token
Outputs	Aggregated transaction metrics (frequency, volume, regularity) stored in farmer profile
Priority	HIGH

FR-H-004: IoT Sensor Data Readiness

Attribute	Detail
ID	FR-H-004
Title	IoT Data Ingestion Architecture
Description	The system architecture shall support future ingestion of IoT sensor data (soil moisture, temperature, rainfall). APIs and data schema shall be IoT-ready.
Priority	MEDIUM (future-ready)

FR-H-005: Climate and Weather Data Ingestion

Attribute	Detail
ID	FR-H-005
Title	Climate Resilience Data
Description	The system shall ingest publicly available climate and weather data relevant to registered farm locations to assess climate risk exposure.
Inputs	Farm GPS coordinates, historical and forecast weather data (external API)
Outputs	Climate risk score component stored per farm
Priority	MEDIUM

MODULE 2: THE BRAIN — ML Intelligence Engine

FR-B-001: Agricultural Credit Score Generation

Attribute	Detail
ID	FR-B-001
Title	Credit Score Calculation
Description	The system shall compute an agricultural credit score for each registered farmer based on a weighted combination of geospatial, transactional, and alternative data inputs.
Inputs	NDVI time-series, transaction history, mobile money data, climate resilience index
Outputs	Numerical credit score (defined range, e.g., 300–850) with confidence rating
Update Frequency	Score recalculated upon new satellite data ingestion or significant data event
Priority	HIGH

Credit Score Input Weights (Initial Model — Subject to Tuning):

Data Stream	Category	Weight (Indicative)
NDVI / Satellite Health	Geospatial	~35%
Sales & Payment History	Transactional	~35%
Mobile Money Activity	Alternative	~20%
Climate Resilience Index	Alternative	~10%

FR-B-002: Yield Prediction

Attribute	Detail
ID	FR-B-002
Title	Crop Yield Forecasting
Description	The system shall generate yield predictions for registered farm parcels for the current growing season based on satellite imagery trends, historical yield data, and climate inputs.
Inputs	Time-series NDVI values, historical yield records, climate data, crop type
Outputs	Predicted yield (quantity per hectare) with confidence interval
Priority	HIGH

FR-B-003: Risk Classification

Attribute	Detail
ID	FR-B-003
Title	Loan Risk Classification
Description	The system shall classify farmers into risk tiers (e.g., Low / Medium / High) to support bank lending decisions, based on the computed credit score and supplementary risk factors.
Outputs	Risk tier label, contributing risk factors, recommended loan range
Priority	HIGH

FR-B-004: Model Retraining Pipeline

Attribute	Detail
ID	FR-B-004
Title	Continuous Model Improvement
Description	The system shall support a model retraining pipeline that incorporates new labeled data (actual vs. predicted yields, loan repayment outcomes) after each harvest season to improve model accuracy over time.
Priority	MEDIUM

FR-B-005: Regional Heatmap Generation

Attribute	Detail
ID	FR-B-005
Title	Risk and Productivity Heatmaps
Description	The system shall generate geospatial heatmaps aggregating credit risk scores and productivity metrics across geographic regions for display in the bank dashboard.
Outputs	GeoJSON or tile-based map layer with risk/productivity classification
Priority	HIGH

MODULE 3: THE MARKETPLACE — User Interfaces

3A. Farmer Mobile Application

FR-M-001: Farmer Registration and Digital Profile Creation

Attribute	Detail
ID	FR-M-001
Title	Farmer Onboarding and Profile Setup
Description	The mobile application shall enable new farmers to register by providing personal details, farm location (GPS), crop types, farm size, and linking their national digital ID and mobile money account.
Inputs	Name, national ID number, phone number, farm GPS coordinates, crop type(s), farm size (hectares)
Outputs	Verified farmer profile created in the system
Priority	HIGH

FR-M-002: Credit Score Dashboard (Farmer View)

Attribute	Detail
ID	FR-M-002
Title	Farmer Credit Score Display

Description	The mobile app shall display a farmer's current credit score, score history over time, and actionable insights on how to improve the score.
Outputs	Score display, trend chart, improvement tips
Priority	HIGH

FR-M-003: Crop Health Monitoring View

Attribute	Detail
ID	FR-M-003
Title	Farm Health Dashboard
Description	The mobile app shall display NDVI-based crop health status for the farmer's registered parcels, with a visual health indicator (e.g., color-coded health level) and a historical trend.
Priority	HIGH

FR-M-004: SMS Alerts

Attribute	Detail
ID	FR-M-004
Title	Crop Health and Loan Status SMS Notifications
Description	The system shall send SMS alerts to farmers for: (a) significant drops in NDVI / crop health, (b) loan application status updates, and (c) upcoming repayment reminders.
Trigger Conditions	NDVI drop > threshold; loan status change; repayment due within X days
Priority	HIGH

FR-M-005: Loan Application

Attribute	Detail
ID	FR-M-005
Title	Loan Application Submission
Description	Farmers shall be able to submit a loan application through the mobile app, selecting a loan amount and purpose. The system shall display the farmer's credit score and estimated eligibility at the time of application.
Priority	HIGH

FR-M-006: Yield Insights Display

Attribute	Detail
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ID	FR-M-006
Title	Yield Forecast and Agricultural Insights
Description	The mobile app shall display the AI-generated yield forecast for the current season and provide actionable agronomic recommendations to improve yield (e.g., optimal planting windows, pest alerts).
Priority	MEDIUM

FR-M-007: Crop Calendar

Attribute	Detail
ID	FR-M-007
Title	Crop Calendar Feature
Description	The mobile app shall provide a crop calendar feature tailored to the farmer's registered crop types and geographic region, indicating key planting, growing, and harvest milestones.
Priority	MEDIUM

3B. Bank / Financial Institution Web Dashboard

FR-B-D-001: Bank Onboarding and Authentication

Attribute	Detail
ID	FR-B-D-001
Title	Bank User Account Management
Description	The web dashboard shall support secure registration and authentication for bank partner users, with role-based access control (e.g., Viewer, Analyst, Administrator).
Priority	HIGH

FR-B-D-002: Farmer Credit Report Access

Attribute	Detail
ID	FR-B-D-002
Title	Individual Farmer Credit Report
Description	Bank users shall be able to search for and retrieve the full credit report for any farmer who has consented to data sharing. The report shall include credit score, risk tier, data contributors, yield forecast, and score trend.
Priority	HIGH

FR-B-D-003: Regional Risk Heatmap

Attribute	Detail
ID	FR-B-D-003
Title	Interactive Risk and Productivity Heatmap
Description	The dashboard shall display an interactive geospatial heatmap showing credit risk distribution and agricultural productivity across regions. Bank users can filter by crop type, risk tier, and time period.
Priority	HIGH

FR-B-D-004: Loan Portfolio Monitoring

Attribute	Detail
ID	FR-B-D-004
Title	Active Loan Portfolio Dashboard
Description	The dashboard shall allow banks to monitor the performance of all active loans disbursed through the platform, including repayment status, score changes in the loan period, and early warning flags.
Priority	HIGH

FR-B-D-005: Automated Credit Scoring API

Attribute	Detail
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ID	FR-B-D-005
Title	Credit Scoring API for Bank System Integration
Description	The system shall expose a REST API endpoint that allows partner banks to programmatically query a farmer's credit score for integration into their own loan origination systems.
Inputs	Farmer unique identifier, API key
Outputs	JSON response with credit score, risk tier, and data timestamp
Priority	HIGH

FR-B-D-006: Loan Application Review

Attribute	Detail
ID	FR-B-D-006
Title	Loan Application Management
Description	Bank users shall be able to view, approve, or reject loan applications submitted through the platform, with full credit data context available during review.
Priority	HIGH

3C. Admin Portal

FR-A-001: User and Partner Management

Attribute	Detail
ID	FR-A-001
Title	Platform Administration
Description	Agri-Lend administrators shall have access to a portal for managing farmer profiles, bank partner accounts, data pipeline status, and system health monitoring.
Priority	HIGH

FR-A-002: Data Pipeline Monitoring

Attribute	Detail
ID	FR-A-002
Title	Data Ingestion Pipeline Monitoring
Description	The admin portal shall display real-time status of all data ingestion pipelines (satellite fetch jobs, mobile money syncs, etc.), including success/failure rates and last run timestamps.
Priority	HIGH

FR-A-003: ML Model Performance Dashboard

Attribute	Detail
ID	FR-A-003
Title	Model Monitoring and Performance Tracking
Description	The admin portal shall display key ML model performance metrics (accuracy, precision, recall, drift indicators) and flag when model retraining is recommended.
Priority	MEDIUM

MODULE 4: Cross-Cutting Features

FR-X-001: National Digital ID Integration

Attribute	Detail
ID	FR-X-001
Title	Digital Identity Verification
Description	The system shall integrate with national digital ID APIs to verify farmer identities during onboarding and link digital ID records to farmer profiles.
Priority	HIGH

FR-X-002: Consent Management

Attribute	Detail
ID	FR-X-002
Title	Data Sharing Consent
Description	The system shall implement an explicit, informed consent mechanism for all data sharing. Farmers must provide documented consent before their data is shared with financial institutions or used for credit scoring. Consent can be revoked at any time.
Priority	HIGH (Compliance Critical)

FR-X-003: Data Insights Reports (Government / NGO)

Attribute	Detail
ID	FR-X-003
Title	Aggregated Regional Productivity Reports
Description	The system shall be capable of generating anonymized, aggregated reports on regional agricultural productivity, lending activity, and financial inclusion metrics for Government and NGO clients.
Outputs	PDF / CSV reports; optionally delivered via API
Priority	MEDIUM

FR-X-004: Input Marketplace (Phase 4)

Attribute	Detail
ID	FR-X-004
Title	Agricultural Input Marketplace
Description	In Phase 4, the platform shall incorporate a marketplace module allowing farmers to discover and purchase agricultural inputs (seeds, fertilizers, tools) from partner suppliers, with the option to finance purchases through integrated loan facilities.
Priority	LOW (Future Phase)

5. Non-Functional Requirements 5.1 Performance

ID	Requirement
NFR-P-001	The credit score API shall return a response within 3 seconds under normal load conditions
NFR-P-002	The system shall support at minimum 10,000 concurrent registered farmer profiles at launch, scaling to 1,000,000+ over time
NFR-P-003	The bank dashboard shall load within 5 seconds on a standard broadband connection
NFR-P-004	Satellite data ingestion jobs shall complete processing for all registered farms within 24 hours of new imagery availability
NFR-P-005	The farmer mobile app shall function acceptably on 2G/3G network conditions with optimized data payloads

5.2 Scalability

ID	Requirement
NFR-S-001	The backend API shall be horizontally scalable to handle increased load without architectural changes
NFR-S-002	The database shall support geospatial indexing via PostGIS to efficiently query farm polygon data at scale
NFR-S-003	The ML model serving infrastructure shall be deployable in a containerized, auto-scaling configuration

5.3 Security

ID	Requirement
NFR-SEC-001	All data in transit shall be encrypted using TLS 1.2 or higher
NFR-SEC-002	All data at rest (database, file storage) shall be encrypted using AES-256

NFR-SEC-003	API authentication shall use OAuth 2.0 / JWT tokens with configurable expiration
NFR-SEC-004	Role-Based Access Control (RBAC) shall be enforced at all API endpoints
NFR-SEC-005	The system shall implement rate limiting on all public-facing API endpoints to prevent abuse
NFR-SEC-006	All user passwords shall be hashed using bcrypt or equivalent strong hashing algorithm
NFR-SEC-007	Personally identifiable information (PII) access shall be logged for audit trail purposes

5.4 Reliability and Availability

ID	Requirement
NFR-R-001	The core platform (API and database) shall target 99.5% uptime (approximately 44 hours downtime per year)
NFR-R-002	The system shall implement automated database backups with a Recovery Point Objective (RPO) of 24 hours and a Recovery Time Objective (RTO) of 4 hours
NFR-R-003	Satellite data pipeline failures shall generate automated alerts to administrators and shall not interrupt farmer or bank user access to the platform

5.5 Usability

ID	Requirement
NFR-U-001	The farmer mobile app shall support multilingual interfaces , including local languages relevant to target deployment regions
NFR-U-002	The farmer mobile app UI shall be designed for users with low digital literacy , utilizing icons, visual indicators, and minimal text input
NFR-U-003	The bank dashboard shall conform to standard WCAG 2.1 Level AA accessibility guidelines
NFR-U-004	The mobile app shall function in an offline-first mode for core features (profile view, score display), synchronizing when connectivity is restored

5.6 Maintainability

ID	Requirement
NFR-M-001	All backend API endpoints shall be documented using OpenAPI / Swagger specification
NFR-M-002	The codebase shall maintain a minimum unit test coverage of 80% for core business logic modules
NFR-M-003	The ML models shall be versioned with the ability to roll back to a previous model version within 1 hour

5.7 Compliance

ID	Requirement
NFR-C-001	The platform shall comply with applicable data protection regulations in all operating countries (e.g., GDPR principles, local equivalents)
NFR-C-002	All financial data processing shall comply with relevant central bank regulations and financial data handling guidelines in operating jurisdictions
NFR-C-003	Credit scoring algorithms shall be designed to be explainable — the system shall be able to provide human-readable justification for any credit score produced

6. External Interface Requirements 6.1 External API Integrations

Integration	Type	Purpose
Google Earth Engine	REST API	Satellite imagery retrieval and NDVI computation
Sentinel-2 (ESA Copernicus)	Satellite Data API	Raw satellite data source
National Digital ID System	REST API	Farmer identity verification during onboarding
Mobile Money Provider(s)	REST API	Transaction history retrieval (with consent) and loan disbursement (Phase 3)

SMS Gateway (e.g., Africa's Talking, Twilio)	REST API	Sending SMS alerts to farmers
Weather / Climate Data API	REST API	Climate risk data ingestion

6.2 Bank Integration Interface

- Agri-Lend shall expose a **REST API** for partner bank integration
- API shall use **JSON** as the data exchange format
- Authentication via **API keys** and **OAuth 2.0**
- Versioned endpoints (e.g., `/api/v1/`) to ensure backward compatibility
- Full **API documentation** provided via Swagger UI

6.3 Hardware Interfaces

- The farmer mobile application shall target **Android** devices (primary), with iOS as a secondary target
- Minimum device requirement: Android 8.0+, 1GB RAM, 100MB storage
- IoT sensor interfaces shall be documented as a future integration point

7. Data Requirements

7.1 Core Data Entities Farmer Profile

1. Farmer Profile

Field	Type / Constraint
farmer_id	UUID (Primary Key)
national_id	String (Unique)
full_name	String
phone_number	String (Unique)
mobile_money_id	String (Optional)
registration_date	DateTime
consent_status	Boolean
consent_date	DateTime

2. Farm Parcel

Field	Type / Constraint
parcel_id	UUID (Primary Key)
farmer_id	Foreign Key → FarmerProfile
parcel_name	String
location_polygon	GeoJSON / PostGIS Geometry
size_hectares	Decimal
primary_crop	String
region	String
created_at	DateTime

3. Credit Score Record

Field	Type / Constraint
score_id	UUID (Primary Key)
farmer_id	Foreign Key → FarmerProfile
score_value	Integer (300–850)
risk_tier	Enum (LOW / MEDIUM / HIGH)
geospatial_score	Decimal
transactional_score	Decimal
alternative_score	Decimal
model_version	String
calculated_at	DateTime
confidence_rating	Decimal

4. Satellite Observation

Field	Type / Constraint
observation_id	UUID (Primary Key)
parcel_id	Foreign Key → FarmParcel
observation_date	Date
ndvi_value	Decimal
cloud_cover_pct	Decimal
data_source	String ("Sentinel-2")
processed_at	DateTime

5. Loan Application

Field	Type / Constraint
application_id	UUID (Primary Key)
farmer_id	Foreign Key → FarmerProfile
bank_id	Foreign Key → BankPartner
requested_amount	Decimal
loan_purpose	String
credit_score_at_application	Integer
status	Enum (PENDING / APPROVED / REJECTED / DISBURSED)
submitted_at	DateTime
reviewed_at	DateTime
reviewed_by	Foreign Key → BankUser

6. Bank Partner

Field	Type / Constraint
bank_id	UUID (Primary Key)
bank_name	String
api_key	String (Hashed)
subscription_tier	String
onboarding_date	DateTime
is_active	Boolean

7.2 Data Retention Policy

Data Type	Retention Period
Farmer PII	Duration of active account + 7 years (regulatory)
Credit Score History	10 years
Satellite Observations	5 years
Transaction Records	7 years
Audit Logs	3 years
Anonymized Aggregate Data	Indefinite

8. System Constraints

8.1 Technical Constraints

- The backend must be developed using **Python and FastAPI**
- The primary database must be **PostgreSQL** with **PostGIS** extension for geospatial support
- The web frontend must be developed in **React (Vite)**

- Satellite data processing must use **Google Earth Engine** API and/or **Sentinel-2** data sources

8.2 Business Constraints

- The platform **does not disburse loans directly** — it is a TSP and intelligence layer
- Initial pilot scope is limited to **Coffee and Teff** value chains
- Initial geographic focus is on **high-production clusters** in the target country/region
- The platform must maintain **data sovereignty** compliance — farmer data shall not leave the jurisdiction without explicit consent and legal basis

8.3 Regulatory Constraints

- Credit scoring models must be **auditable and explainable** to comply with potential consumer credit regulations
- Farmer data sharing with banks requires **explicit opt-in consent**
- Financial data handling must conform to any applicable central bank or financial regulatory body guidelines

9. Appendix 9.1 Phased Delivery Roadmap

Phase	Name	Key Deliverables
Phase 1	Foundation	Farmer onboarding, digital ID integration, farm profile creation, basic mobile app
Phase 2	Monitoring	Satellite NDVI ingestion, crop health dashboard, SMS alerts, initial credit score model
Phase 3	Fintech	Mobile money integration, loan application workflow, bank dashboard, credit scoring API
Phase 4	Ecosystem	Agricultural input marketplace, expanded crop types, NGO/Government data reports

9.2 Revenue Model Context (Non-Functional Reference)

The following business model context informs feature priority decisions:

Revenue Stream	Model	Implication for System

SaaS Licensing	Banks pay monthly subscription fees	Bank dashboard and scoring API are high-priority deliverables
Transaction Fees	Commission per loan disbursed via platform	Loan application tracking and disbursement integration are required
Data Insights	Reports sold to Government/NGOs	Aggregated reporting module required

9.3 Impact Metrics (Aspirational Targets)

Metric	Target
Financial Inclusion	85% improvement in credit access for registered farmers
Productivity Gain	60% improvement in yield for farmers using platform insights
Food Security Impact	75% positive contribution to food security indicators

This SRS is a living document. It should be reviewed and updated as new requirements emerge from stakeholder consultation, pilot feedback, and regulatory guidance. All requirements should be traced to specific development tickets in the project management system.

End of Document